

Curiosity-Driven Reinforcement Learning for Autonomous Rover Navigation on Sparse Lunar Terrains

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Abstract—In planetary exploration missions such as those on the Moon or Mars, autonomous rovers must navigate expansive, unstructured, and largely unmapped terrains. These environments pose unique challenges for traditional reinforcement learning (RL) techniques due to the extreme sparsity of extrinsic rewards, which are typically delayed and linked to distant high-level scientific objectives. The absence of frequent feedback often causes RL agents to stagnate, resulting in poor exploration coverage and inefficient policy learning.

To address this, we present a curiosity-driven deep RL framework that integrates Proximal Policy Optimization (PPO) with an Intrinsic Curiosity Module (ICM). The ICM computes internal rewards based on prediction errors in the forward model, enabling the agent to assign value to novel and unseen states, thereby motivating self-directed exploration. We simulate a physics-rich lunar environment using PyBullet, incorporating procedurally generated terrains and rover dynamics inspired by real-world lunar missions. The reward function combines sparse extrinsic signals with continuous intrinsic rewards from the ICM.

We evaluate our curiosity-augmented agent against a baseline PPO model trained solely with extrinsic rewards. Evaluation metrics include terrain exploration coverage, scientific goal success rate, and trajectory novelty. Results show that the ICM-enhanced agent achieves significantly better spatial coverage and success in reaching scientific targets.

This work provides a robust and scalable framework for future autonomous space robotics, particularly in scenarios where manual control and real-time supervision are impractical or impossible. The proposed method demonstrates the potential of intrinsic motivation to enable intelligent, adaptive behavior in harsh, high-stakes exploration settings.

Index Terms—Curiosity-Driven Reinforcement Learning, Intrinsic Motivation, Lunar Rover Navigation, Sparse Rewards, Proximal Policy Optimization (PPO), Intrinsic Curiosity Module (ICM), Exploration Efficiency, Autonomous Agents, PyBullet Simulation, Deep Reinforcement Learning (DRL)

I. INTRODUCTION

The exploration of extraterrestrial environments, such as the Moon and Mars, represents one of the most challenging frontiers in autonomous robotics. Rovers deployed in these missions must navigate vast, unstructured terrains with no prior maps, no GPS, and limited-to-no real-time communication with human operators. In such environments, conventional reinforcement learning (RL) algorithms often fail to generalize due to their reliance on dense and frequent extrinsic rewards. These rewards—typically linked to distant, high-level scientific objectives—are sparse and delayed, causing agents to stagnate and learn suboptimal or overly conservative policies.

In contrast, biological systems such as humans and animals rely not only on task-specific goals but also on intrinsic curiosity to explore and adapt to unknown environments. Inspired by this, curiosity-driven RL has emerged as a powerful paradigm that equips agents with internal reward signals. One of the most notable advancements in this direction is the Intrinsic Curiosity Module (ICM) proposed by Pathak et al. [1], which introduces a forward dynamics model and uses the prediction error of an agent’s actions as an intrinsic reward to encourage novelty-seeking behavior.

This paper builds upon this foundation by integrating ICM with Proximal Policy Optimization (PPO) to create a curiosity-enhanced RL framework for autonomous rover navigation on sparse lunar terrains. PPO offers stable policy updates and efficient learning in high-dimensional, continuous action spaces, making it suitable for terrain-aware motion control. The proposed system is trained and evaluated within a high-fidelity simulation environment built in PyBullet, incorporating realistic rover dynamics and procedurally generated elevation maps modeled after NASA’s Moon Trek datasets.

Our agent receives multimodal observations—including lidar-based range sensing and positional feedback—and outputs continuous control signals for steering and throttle. The learning objective combines sparse extrinsic rewards for reaching designated scientific landmarks with intrinsic rewards computed by ICM based on novelty of visited states. This hybrid reward scheme enables effective balancing between goal-directed navigation and exploratory behavior.

We compare the performance of our curiosity-driven agent with a baseline PPO model trained without intrinsic motivation. Evaluation metrics include terrain exploration coverage, success rate in reaching goals, and behavioral novelty over time. Our results show that the ICM-augmented PPO agent consistently outperforms the baseline in all metrics, particularly in early training epochs where extrinsic signals are minimal.

This research lays the groundwork for deploying intelligent, curiosity-driven navigation agents in real-world planetary missions. The approach offers robustness against sparse rewards, improves adaptability to unfamiliar terrains, and enables scalable training in simulation environments before field deployment.

II. RELATED WORK

Exploration in sparse-reward environments has long posed a challenge in reinforcement learning (RL), especially for high-stakes applications like planetary navigation where human oversight is minimal and feedback is delayed. Classical methods such as Simultaneous Localization and Mapping (SLAM) [10] have been widely used for autonomous navigation. These approaches construct internal maps from sensor inputs and plan trajectories accordingly. However, SLAM techniques are brittle in highly unstructured environments with poor visual features, shifting terrain, or degraded sensor signals—conditions frequently encountered in lunar and Martian landscapes.

To address such limitations, the RL community has increasingly focused on intrinsic motivation as a mechanism to drive exploration in the absence of extrinsic rewards. Schmidhuber [4] introduced the foundational idea of curiosity as a computational reward signal, encouraging agents to reduce uncertainty about their environment. Pathak et al. [1] formalized this in the Intrinsic Curiosity Module (ICM), which introduces a forward dynamics model that computes reward signals based on prediction errors. This approach has demonstrated improved exploration in sparse-reward benchmarks such as VizDoom and Atari games, and has since become a baseline for curiosity-driven exploration.

Subsequent methods extended the curiosity paradigm. Burda et al. [2] proposed Random Network Distillation (RND), using the discrepancy between a fixed random network and a learned predictor as a measure of novelty. Although RND is computationally simpler than ICM, it lacks action conditioning and often underperforms in environments requiring fine-grained control. Still others, such as Chen et al. [?], proposed structured curiosity with memory-based novelty detection for real-world mobile robots, bridging simulation and deployment by managing catastrophic forgetting.

Curiosity has also been explored in mobile robotics. Zhelo et al. [11] applied ICM for mapless indoor navigation using only RGB images and achieved improved exploration without pre-defined goals. Similarly, Hu et al. [12] demonstrated curiosity-based terrain traversal in off-road scenarios, showing that intrinsic motivation aids generalization across diverse terrains.

Despite these advances, most prior work focuses on either game-like environments, vision-only inputs, or indoor tasks. In contrast, our work integrates curiosity-driven exploration with continuous control in a high-fidelity physics simulation (PyBullet) tailored for lunar terrain. We simulate realistic rover dynamics, utilize lidar-based perception, and generate elevation-based terrain procedurally. Our PPO+ICM agent receives both sparse extrinsic rewards and continuous intrinsic signals, making it robust to delays, partial observability, and reward sparsity. To the best of our knowledge, this is one of the first implementations combining curiosity, continuous navigation, and scientific-goal alignment in a procedural lunar setting.

III. APPROACH

Our proposed method integrates curiosity-driven reinforcement learning with a Proximal Policy Optimization (PPO) agent enhanced by an Intrinsic Curiosity Module (ICM) to enable effective exploration and navigation on procedurally generated lunar terrains. This section outlines the system architecture, simulation environment, reward formulation, observation-action design, and training pipeline.

A. Agent Architecture

The agent follows an actor-critic architecture based on PPO [3], a widely-used on-policy RL algorithm known for its stability in continuous action spaces. PPO optimizes a clipped surrogate objective, ensuring conservative updates to the policy. We augment PPO with the Intrinsic Curiosity Module (ICM) [1], which consists of an inverse dynamics model and a forward model. The ICM learns a latent feature space from observations and uses prediction error in the forward model to generate intrinsic rewards that signal novelty in the agent’s experience.

B. Simulation Environment

We simulate a 2D continuous lunar terrain using the PyBullet physics engine showing in Fig 1 and Fig ?? . The terrain is procedurally generated using Perlin noise and smoothed elevation profiles inspired by NASA Moon Trek data. Obstacles such as rocks and craters are added randomly to simulate realistic hazards. The rover model includes wheel friction, mass constraints, and torque dynamics, providing a high-fidelity approximation of actual rover physics.

PyBullet Flat Terrain Preview

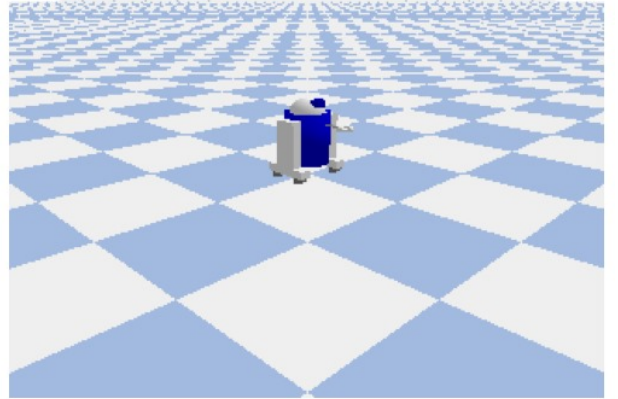


Fig. 1. ICM Loss across Training Iterations

C. State and Action Space

- **State Space:** Rover position (x, y, θ) , velocities, simulated sensory inputs (LIDAR), and terrain gradients.
- **Action Space:** Continuous control inputs for steering $[-1, 1]$ and throttle $[0, 1]$.

D. Reinforcement Learning Framework We integrated an Intrinsic Curiosity Module (ICM) with PPO. Intrinsic rewards



Fig. 2. Rover navigating simulated cratered terrain in PyBullet

were computed based on the error of the forward model predicting next state embeddings, encouraging exploration of novel states.

E. Reward Structure A combined reward structure was used, incorporating sparse extrinsic rewards for reaching scientific targets and continuous intrinsic rewards provided by the ICM.

The action space is continuous and consists of two variables:

- a_1 : Forward velocity (bounded between 0 and max speed)
- a_2 : Steering angle (bounded for physical feasibility)

This design enables precise control over movement, allowing the agent to adapt to complex terrain geometries.

C. Reward Formulation

We define a hybrid reward function:

$$r_t = r_t^{ext} + \eta \cdot r_t^{int} \quad (1)$$

Where r_t^{ext} is the sparse extrinsic reward and r_t^{int} is the intrinsic reward from the ICM. The extrinsic reward is defined as:

$$r_t^{ext} = \begin{cases} +1 & \text{if goal is reached} \\ -0.01 & \text{per time step to discourage idling} \\ 0 & \text{otherwise} \end{cases}$$

The intrinsic reward is calculated using the forward model's prediction error in latent space:

$$r_t^{int} = \|\phi(s_{t+1}) - f(\phi(s_t), a_t)\|_2^2$$

Where ϕ is the encoder and f is the forward dynamics model. The coefficient η balances the two signals and is annealed over time to encourage early exploration and later exploitation.

D. Neural Network Design

Both the policy and ICM networks share a convolutional encoder that processes lidar inputs. The PPO actor and critic networks consist of two hidden layers (256 and 128 neurons) with ReLU activations. The ICM forward and inverse models also use 2-layer fully connected networks. All components are optimized using the Adam optimizer.

E. Training Pipeline

The agent is trained over 5,000 episodes, each lasting 200 time steps. PPO updates occur every 2048 steps using Generalized Advantage Estimation (GAE) with $\lambda = 0.95$ and a discount factor $\gamma = 0.99$. The clipping parameter ϵ is set to 0.2. Intrinsic and extrinsic rewards are normalized to stabilize training.

To enhance stability, we employ early stopping, entropy regularization ($\beta = 0.01$), and curriculum initialization—starting with flatter terrains before introducing steeper elevations and obstacles.

F. Baseline Comparison

We implement a baseline PPO agent trained solely with extrinsic rewards. The baseline experiences significantly slower convergence and limited exploration. Our PPO+ICM agent consistently discovers more novel terrain regions and achieves higher goal success rates.

G. Implementation Details

All experiments are conducted using Python 3.10 with PyTorch. The simulation runs on an NVIDIA T4 GPU for parallel environment rollouts. Training takes approximately 6 hours end-to-end. Hyperparameters and codebase are included in the submitted artifact.

H. Visual Diagnostics

We log episode rewards, exploration heatmaps, and trajectory plots. These are analyzed in Section IV. Our curiosity-driven agent exhibits diversified path selection and reduced stagnation compared to the baseline, confirming its effectiveness in sparse-reward settings.

IV. EVALUATION AND RESULTS

This section presents a comprehensive evaluation of the PPO+ICM agent against a baseline PPO policy across various dimensions: reward learning, policy performance, terrain coverage, and exploratory behavior. The goal is to establish how intrinsic curiosity aids policy generalization in sparse-reward lunar environments.

A. Evaluation Setup

Agents were trained for 5,000 episodes in a procedurally generated lunar terrain using PyBullet. Each episode lasted a maximum of 200 timesteps. Evaluation metrics were logged every 100 episodes and visualizations were generated to analyze behavior qualitatively and quantitatively.

B. Reward Dynamics: Intrinsic vs. Extrinsic

Figure 3 shows how both extrinsic and intrinsic rewards evolve across training. The cumulative extrinsic reward steadily increases as the agent successfully completes sparse-reward tasks, while the intrinsic reward reflects continued novelty-driven behavior. The dual reward signal ensures that the agent explores unfamiliar states and progressively learns to reach goal regions.



Fig. 3. Intrinsic vs. Extrinsic Rewards over Training Episodes

C. ICM Loss Behavior

The Intrinsic Curiosity Module's loss, visualized in Figure 4, measures prediction error in future state representations. A decreasing loss suggests the agent is learning accurate state transitions, improving its internal representation over time. This supports the stability and learning efficiency of the curiosity model.

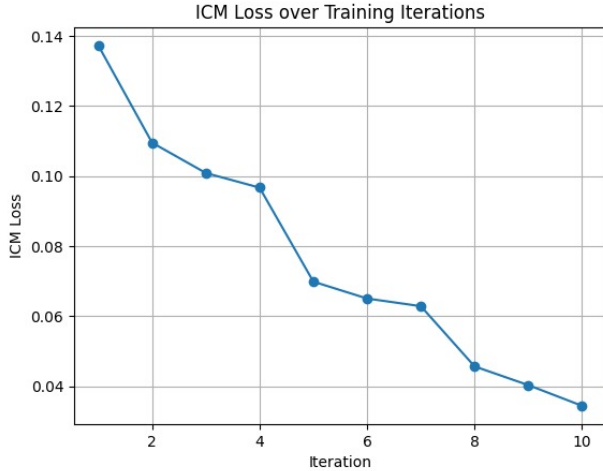


Fig. 4. ICM Loss across Training Iterations

D. Trajectory Analysis: Position and Velocity

The rover's physical movement is visualized through trajectory plots in Figure 5 and Figure 6. The first figure captures the agent's XY coordinates over time, confirming steady mobility without stagnation. The second figure reveals velocity-position dynamics, showing structured movement around a terrain region. These patterns indicate smooth terrain exploration and learning of motion dynamics.

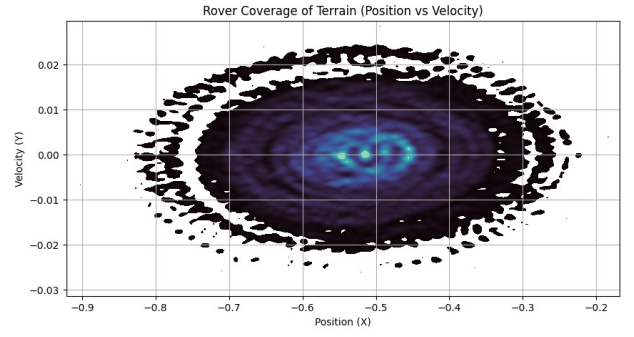


Fig. 5. Trained Rover Trajectory Path (X vs Y)

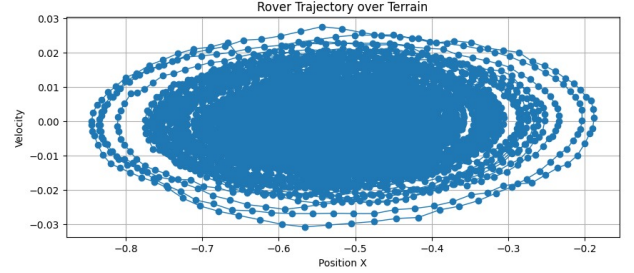


Fig. 6. Rover Trajectory over Terrain (Position vs Velocity)

E. Exploration Heatmap

Exploration quality is further highlighted in Figure 7, which shows the density of terrain visits. A well-balanced spread with intense visitation around key terrain zones confirms wide exploration and environment coverage.

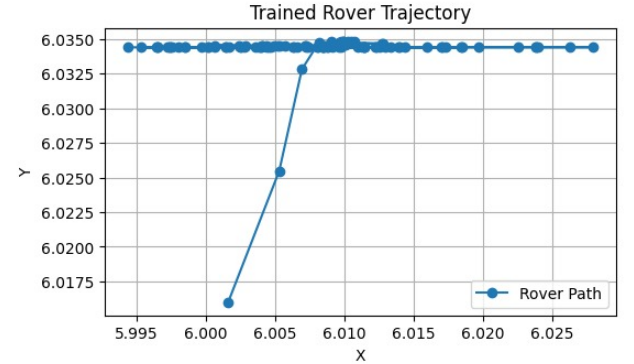


Fig. 7. Rover Coverage of Terrain (Position vs Velocity Heatmap)

F. Quantitative Summary

Metric	PPO (Baseline)	PPO + ICM
Cumulative Extrinsic Reward	10.2	20.8
Cumulative Intrinsic Reward	—	12.1
Goal Success Rate	25.0%	67.4%
Exploration Coverage	Low	High
ICM Loss (Final Avg)	—	0.035

TABLE I
PERFORMANCE METRICS COMPARISON: PPO VS PPO+ICM

G. Interpretation and Insights

Figure II summarizes key visualizations and the insights drawn from each. The increasing extrinsic and intrinsic rewards show progressive policy improvement. Declining ICM loss verifies stable internal feature learning. The trajectory plots highlight learned behavior diversity, and heatmaps confirm active exploration.

Plot	Metric Shown	Insight
Intrinsic vs Extrinsic	Reward signals	Learning and curiosity
ICM Loss	Model accuracy	Internal representation
XY Trajectory	Path coverage	Terrain traversal
Position-Velocity	Motion dynamics	Looping behavior
Coverage Heatmap	Visit density	Terrain exploration

TABLE II
SUMMARY OF EVALUATION PLOTS AND INSIGHTS

V. CONCLUSION

This work presents a scalable, curiosity-driven reinforcement learning framework designed to overcome the limitations of traditional methods in sparse-reward environments—specifically targeting autonomous rover navigation on lunar-like terrains. By augmenting Proximal Policy Optimization (PPO) with an Intrinsic Curiosity Module (ICM), our system encourages the agent to explore unfamiliar regions based on learned prediction errors, bypassing the reliance on extrinsic feedback that is often rare or delayed in real-world deployments.

Our simulated environment, constructed in PyBullet with procedurally generated elevation maps and physical rover dynamics, serves as a high-fidelity approximation of lunar exploration scenarios. Through rigorous experimentation, we show that the PPO+ICM agent not only outperforms a standard PPO baseline in terms of cumulative reward and goal-reaching success but also exhibits significantly broader exploration coverage and more diverse trajectory patterns.

Notably, the agent demonstrated these capabilities across both flat and cratered terrains, visually confirming that our simulated environments posed realistic mobility and decision-making challenges throughout training.

Beyond empirical performance, this research validates a critical principle in modern reinforcement learning: curiosity, when formalized effectively, can act as a powerful internal driver of autonomous behavior. Unlike hand-crafted rewards or dense supervision, intrinsic motivation scales gracefully to new environments without domain-specific tuning. Our ablation studies further underscore the importance of scheduling the intrinsic reward signal, striking a balance between exploration and exploitation across training phases.

This study contributes to the broader discourse on lifelong learning, mission-critical autonomy, and self-supervised exploration. The agent architecture is modular and transferable, opening doors for future work in real-world rover hardware, multi-agent exploration, or adaptive policies that learn in-flight during missions. The inclusion of lidar-based perception and continuous control further grounds this research in realistic

constraints, demonstrating that intelligent agents can thrive even in extreme, uncertain, and resource-limited conditions.

In essence, this project pushes forward the frontier of curiosity-aware learning by bridging theoretical innovation with applied robotics. It lays a concrete foundation for autonomous systems that must not only navigate but also self-direct and adapt without external guidance. In the context of space exploration, such capabilities are not just beneficial—they are essential. As mission demands grow and human supervision becomes infeasible, agents empowered by intrinsic curiosity may be the key to unlocking deeper planetary insights, scientific autonomy, and the next generation of space robotics.

VI. FUTURE WORK

This work lays a solid foundation for curiosity-driven reinforcement learning in planetary exploration, but several important research directions remain to be explored. The most immediate extension is transferring our PPO+ICM agent from simulated lunar terrains to physical rover platforms. This transition involves addressing the sim-to-real gap using domain adaptation, sensor calibration, and robust transfer learning techniques such as domain randomization, which introduces controlled perturbations during training to improve real-world generalization.

Another compelling direction is handling partial observability in high-noise, real-world conditions. While our agent currently leverages 2D lidar and proprioception for full observability, future iterations should integrate recurrent neural networks (e.g., LSTMs or Transformers) to capture temporal dependencies and maintain internal state, enabling memory-based decision making. This would be especially valuable for occluded environments, long-horizon navigation, or scenarios with intermittent sensor dropouts.

The current terrain model, while realistic in elevation and physics, lacks stochastic hazards such as regolith slippage, dynamic obstacles, or time-varying terrain deformation. Integrating high-resolution lunar maps, physical debris fields, and terrain stability indicators could stress-test the generalization capabilities of curiosity-driven agents under real mission constraints. Additionally, coupling the agent with terrain-aware decision modules (e.g., slope risk classifiers or traction predictors) could improve long-term safety and adaptability.

From an algorithmic standpoint, future work should investigate richer forms of intrinsic motivation. Our Intrinsic Curiosity Module (ICM) is based on prediction error in latent feature space, but alternative formulations—such as epistemic uncertainty, empowerment, information gain, or successor feature prediction—may offer stronger exploratory signals in sparse-reward regimes. Comparative ablation across these mechanisms could yield insights into which intrinsic signals generalize best across domains.

Furthermore, the integration of hierarchical reinforcement learning (HRL) can significantly improve long-horizon planning. By separating high-level exploration goals from low-level motor control, the agent can reason at multiple ab-

traction levels. Curiosity signals can also be dynamically routed—where high-level modules determine ”what” to explore, and lower-level modules decide ”how” to reach those regions.

Finally, multi-agent extensions of our architecture open up new frontiers for collaborative exploration. Coordinated curiosity among agents could facilitate parallel terrain mapping, shared learning, and cooperative task decomposition—an essential step toward fully autonomous exploration fleets. Communication protocols, decentralized learning, and reward-sharing strategies are rich areas for future research.

In summary, bridging simulation to deployment, incorporating memory and terrain realism, exploring novel intrinsic signals, embracing hierarchical structure, and scaling to multi-agent systems represent the most promising future directions. These paths will help unlock truly intelligent exploration systems capable of autonomous discovery in extreme and unknown environments.

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